

AI-Based Flood Prediction & Big Data Analytics

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Abstract

This project uses AI and big data technologies for flood prediction and analytics.

Temporal Fusion Transformer (TFT) models were trained for time-series forecasting.

Hadoop and Spark were used for distributed processing of environmental datasets.

Kafka enabled real-time streaming of weather and sensor data.

Interactive dashboards were created using Power BI and Tableau for visualization.

Technologies Used

Technology Stack
TensorFlow, TFT, Hadoop, Spark, Kafka, MongoDB, Power BI

Future Scope

This project can be extended with advanced AI models, cloud deployment, real-time scalability, and improved datasets for better accuracy and performance.

Conclusion

The project demonstrates practical implementation of AI, machine learning, or software technologies to solve real-world problems effectively.